

We acknowledge and pay my respects to the traditional owners and custodians of the land on which we walk, work & live.

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Using Excel for Data

Using real Justice Reinvestment related data to build practical spreadsheet skills

Today's agenda



Module 1 — Getting data into Excel

Downloading CSVs, opening correctly, saving your working file



Module 2 — Cleaning your data

Fixing formats, removing junk, making data consistent



Module 3 — Formulas that matter

SUM, AVERAGE, COUNT, rates per 100,000, percentage change



Module 4 — Sorting and filtering

Finding what you need fast, slicing data by category or year



Module 5 — Charts that communicate

Picking the right chart, formatting it properly, avoiding bad habits



Module 6 — Pivot Tables



Module 7 — Power BI in Excel

Some relevant datasets

ROGS

Report on Government Services

- Performance data for all states and territories
- Education, health, justice, housing
- Annual, comparable across jurisdictions
- Format: Excel tables

Good for: comparing states over time

BOCSAR

NSW Bureau of Crime Statistics

- Crime by offence type and LGA
- Year-by-year incident counts
- Free download, updated quarterly
- Format: CSV (comma-separated)

Good for: filtering, trends, rates

ABS CENSUS 2021

Aboriginal & Torres Strait Islander data

- Population by state and LGA
- Age, education, employment
- ~812,000 people — 3.2% of Australia
- Format: Excel and CSV tables

Good for: population denominators



We'll use BOCSAR as our main working file today. ROGS and ABS data come in during the exercises.

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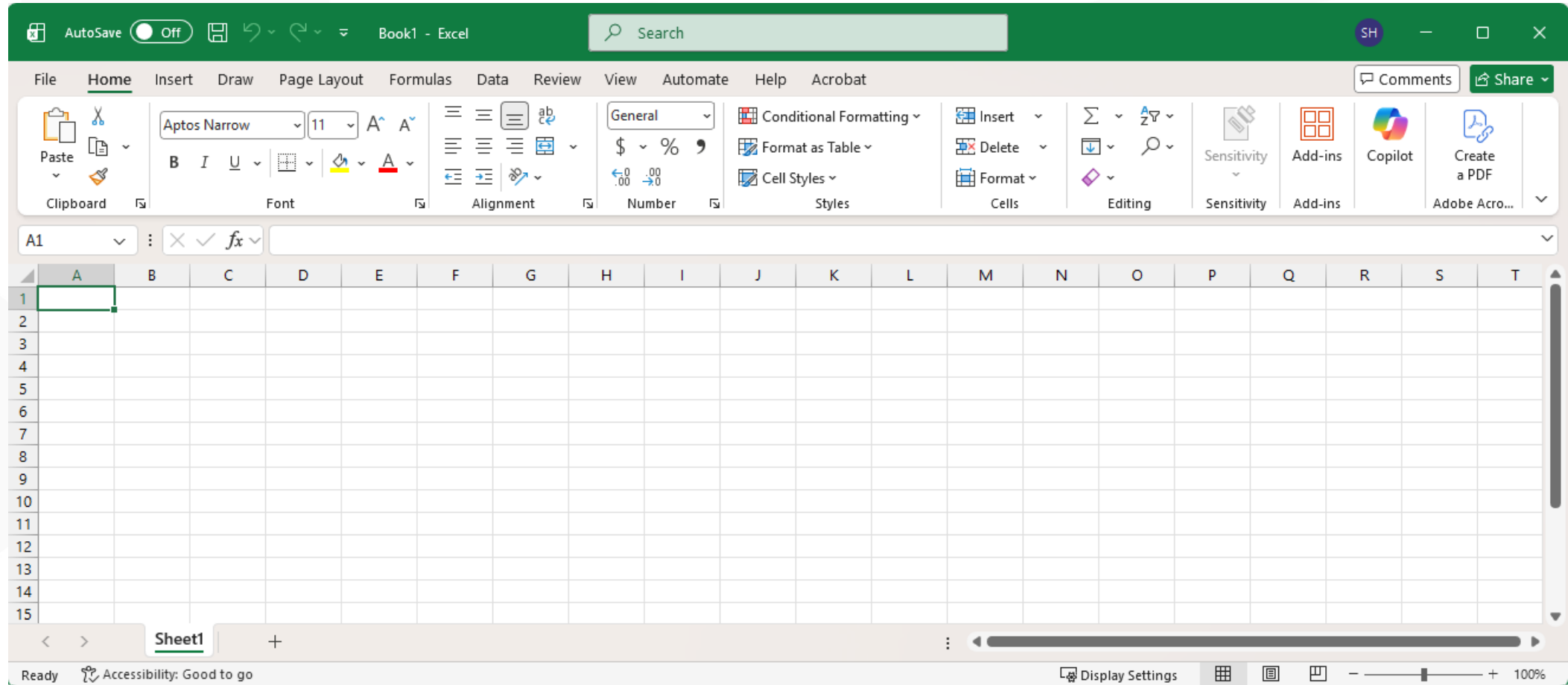
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Excel environment





Module 1 — Getting data into Excel

Downloading CSVs, opening correctly, saving your working file

Getting data into Excel

Government data usually comes as a **CSV file** — plain text with commas separating values. Excel needs a little help reading it correctly.

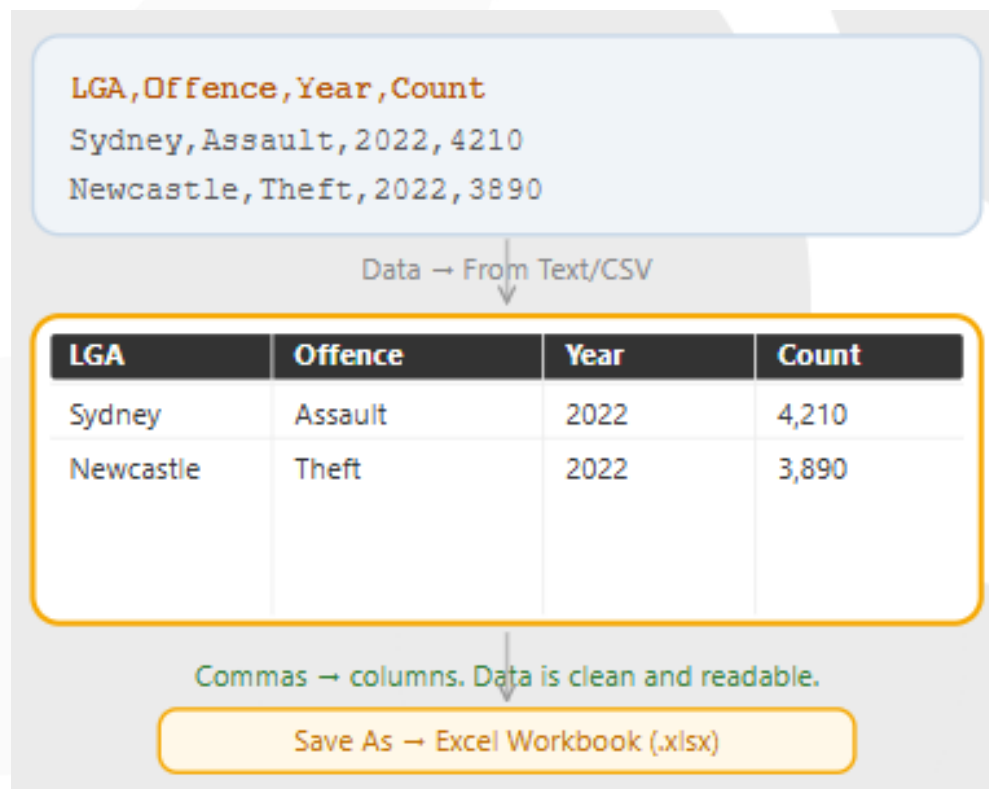
What's a CSV? Open one in Notepad and you'll see rows of text with commas between values. Excel converts each comma into a new column.

Why not just double-click the file?

Excel can guess wrong — dates become numbers, postcodes lose leading zeros

The import wizard gives you full control

Always save a separate **.xlsx working copy** straight away



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LGA, Offence, Year, Count

Sydney, Assault, 2022, 4210

Newcastle, Theft, 2022, 3890

Data → From Text/CSV

LGA	Offence	Year	Count
Sydney	Assault	2022	4,210
Newcastle	Theft	2022	3,890

Commas → columns. Data is clean and readable.

Save As → Excel Workbook (.xlsx)



Downloading and opening the BOCSAR data

- 1 Open your browser and go to `bocsar.nsw.gov.au` → **Statistics & Dashboards** → **Open Datasets**
- 2 Find "**Recorded Crimes — Offences by Local Government Area**" and click the download link. Save the **.csv** file to your Desktop or course folder.
- 3 **Do not double-click the file yet.** Opening directly can cause Excel to misread it. We'll use the import wizard instead.
- 4 Open Excel. Go to `File → Open → Browse`, navigate to your CSV, and click **Open**.
- 5 When prompted, select **Comma** as the delimiter and click **Finish** or **Load**. → *Your data should appear in neat columns*
- 6 Go to `File → Save As`, change the format to **Excel Workbook (.xlsx)**, and save as `BOCSAR_working.xlsx`. The CSV is now your backup — don't touch it.

Take a moment to read the data before touching it

This is a habit every good analyst has — spend 2–3 minutes just **reading** before doing anything else.

- 1 How many **columns** are there? Click the last column header. What does each one represent?
- 2 Press `Ctrl + End` — jumps to the last cell with data. What row number? That's your row count.
- 3 Look at the **first 10 rows**. Are numbers right-aligned? Text left-aligned? Mixed alignment = problem.
- 4 Scroll through the **Offence Type** column. Are names consistent — no typos, no mixed case?

Useful keyboard shortcuts

Ctrl + End — jump to last cell with data
Ctrl + Home — back to cell A1
Ctrl + → — jump to end of a row
Ctrl + ↓ — jump to end of a column
Ctrl + S — save (do this often!)

What to look for

- ✓ Column headers clearly labelled in row 1
- ✓ One piece of information per column
- ✓ Numbers are right-aligned in their cells
- X Blank rows mixed into the data
- X Merged cells anywhere in the data area



Module 2 — Cleaning your data

Fixing formats, removing junk, making data consistent

Cleaning your data — why it matters

Small inconsistencies cause big problems when you sort, filter, or add things up. Fix them first — everything after becomes much easier.

Common problems in real data

- **Extra spaces** — "Sydney " ≠ "Sydney" — filters will miss it
- **Inconsistent capitalisation** — "assault" vs "Assault" vs "ASSAULT"
- **Numbers stored as text** — SUM() returns 0 or ignores them
- **Merged cells** — break sorting and filtering entirely
- **Blank rows** — confuse formulas and pivot tables
- **Mixed date formats** — 01/01/22 vs 1-Jan-22 vs 2022

⚠ **Always work on a copy.** Right-click your sheet tab → **Move or Copy** → tick "Create a copy". Name the original **RAW** — never edit it.

RAW = untouched original — never edit

Working = where you make all changes

Charts = outputs only

RAW 🔒

Working

Charts

Some useful cleaning tools

1. TRIM — REMOVE EXTRA SPACES

```
=TRIM(A2)    "Sydney " → "Sydney"
```

Use in a helper column. Then copy → Paste Special → Values to overwrite the originals.

2. PROPER — FIX CAPITALISATION

```
=PROPER(TRIM(A2))    "ASSAULT" → "Assault"
```

Combining with TRIM fixes both problems in one step.

3. FIND & REPLACE — FIX TYPOS IN BULK

Press Ctrl + H → type what to find → type the correct version → click **Replace All**. Works across the entire sheet instantly.

E.g. replace "Sydney C.B.D." with "Sydney CBD" in one click.

4. CONVERT NUMBERS STORED AS TEXT

Select the column → look for the **green triangle** in the top corner of cells → click the yellow warning icon → choose **Convert to Number**.

It's worked when the numbers shift from left-aligned to right-aligned.

Let's clean the BOCSAR data together

- 1 Make a copy first.** Right-click the sheet tab → Move or Copy → tick "Create a copy". Rename the original **RAW** and the copy **Working**. All edits happen on Working.
- 2 Freeze the header row.** Go to `View → Freeze Panes → Freeze Top Row`. Scroll down — headers should stay visible.
→ *A thin line appears under row 1*
- 3 Check for extra spaces** in the LGA column. In an empty column, type `=TRIM(A2)` and press Enter. Compare the result to the original — are they identical?
- 4 Fix capitalisation.** Press `Ctrl + H`. In "Find what" type a known incorrect value. In "Replace with" type the correct version. Click **Replace All**.
- 5 Check the Count column.** Click a number — is it right-aligned? Try `=SUM()` on the column. If it returns zero, use the Convert to Number option.



Module 3 — Formulas that matter

SUM, AVERAGE, COUNT, rates per 100,000, percentage change

Formulas — the building blocks

Anatomy of a formula

=AVERAGE (D2 : D500)

= starts function name range: D2 to D500

=SUM (D2 : D500) total all values

=AVERAGE (D2 : D500) mean value

=COUNT (D2 : D500) how many numbers

=MAX (D2 : D500) highest value

=MIN (D2 : D500) lowest value

How to enter a formula

1. Click an empty cell
2. Type =
3. Start typing the function name — Excel suggests it
4. Press Tab to accept the suggestion
5. Click and drag to select your range
6. Press Enter

Copying a formula down a column

Click the cell with your formula → hover over the bottom-right corner until you see a small + → double-click it. Excel fills down automatically for all rows.

Rate per 100,000 population

THE MOST IMPORTANT FORMULA IN THIS COURSE

Rate per 100,000 population

Raw counts are almost always misleading when comparing areas. Sydney will always have more crimes than Armidale — because it has far more people. **Rates make comparisons fair.**

$$=(C2 / D2) * 100000$$

C2 = incident count D2 = population

LGA	Thefts	Population	Rate /100k
Sydney CBD	8,920	24,000	37,167
Blacktown	4,100	394,000	1,041
Armidale	380	30,000	1,267

💡 Sydney CBD has *fewer residents* than Blacktown but a much higher rate — millions of workers pass through daily but aren't counted as residents.

Step by step

- 1 Add a **Population** column — use LGA figures from the ABS 2021 Census
- 2 Add a column heading: **Rate per 100k**
- 3 Type `=(C2/D2)*100000` — adjust column letters to match yours
- 4 Double-click the fill handle to copy the formula down the whole column
- 5 Format to **0 decimal places** — rates don't need decimal precision

Exercise: Calculate theft rates for 5 LGAs

- 1 In your Working sheet, identify 5 LGAs you're interested in and note their theft incident counts
- 2 Set up a small table starting at column H with headers: **LGA | Theft count | Population | Rate per 100k**
- 3 Fill in the theft counts. Then go to `abs.gov.au` and look up the 2021 population for each LGA. Enter these in the Population column.
- 4 In the Rate column, type `=(C2/D2)*100000` — adjusting cell references to match your table. Press Enter.
- 5 Copy the formula down. Use `=MAX()` and `=MIN()` on the Rate column. → *Does the ranking change compared to the raw counts?*
- 6 **Write one sentence** in a nearby cell explaining what the rate tells us that the raw count doesn't.



Module 4 — Sorting and filtering

Finding what you need fast, slicing data by category or year



Sorting and filtering — finding what you need

Turning on AutoFilter

- 1 Click anywhere inside your data
- 2 Go to **Data → Filter** (or press `Ctrl + Shift + L`)
- 3 → *Dropdown arrows appear on every column header*
- 4 Click a dropdown → untick **Select All** → tick just what you want → OK
- 5 To clear: click the dropdown → **Clear Filter**. Filtering never deletes data.

Filtering hides rows temporarily — it never deletes them. Your full dataset is always safe.

Sorting

- 1 Click **one cell** in the column you want to sort by — do not select the whole column
- 2 Go to **Data → Sort A→Z** (text) or **Sort Z→A** (largest first for numbers)
- 3 For multiple sort levels: **Data → Sort → Add Level**

⚠ **Never sort a selected column.** Always click a single cell. Selecting and sorting one column disconnects it from the rest of your data.

Exercise: Filter and sort crime data

- 1 Turn on AutoFilter: click in your data → `Data → Filter` . Dropdown arrows appear on the header row.
- 2 Click the dropdown on the **Offence Type** column → untick "Select All" → tick **Assault** → OK. → *The row count at the bottom drops — only assault rows are visible*
- 3 Sort by **Count**, largest to smallest: click a cell in the Count column → `Data → Sort Z→A` . → *Highest-assault LGAs should now be at the top*
- 4 Write down the **top 3 LGAs** by assault count. Are they what you expected?
- 5 Add a second filter on the **Year** column to show the **most recent year only**. Does the ranking change?
- 6 Clear all filters: `Data → Clear` . Confirm all rows are back.



Module 5 — Charts that communicate

Picking the right chart, formatting it properly, avoiding bad habits

Charts — choosing the right one



BAR / COLUMN CHART

Use when **comparing categories** — assault counts by LGA, expenditure by state.

Best for: side-by-side comparisons at one point in time



LINE CHART

Use when showing **change over time** — crime rates per year, expenditure trends.

Best for: trends, patterns, before/after comparisons



PIE CHART — USE SPARINGLY

Only for **parts of a whole** with fewer than 5 categories and clear size differences.

Avoid when slices look similar in size

How to build a chart in Excel

- 1 Select your data** including headers. Hold `Ctrl` to select non-adjacent columns.
- 2** Go to `Insert → Charts`. For comparing LGAs choose **Clustered Bar** — horizontal bars are easier to read when labels are long.
- 3** The chart appears. Click it — the **Chart Design** and **Format** tabs appear in the ribbon.
- 4 Add a title:** click the default title → delete it → type e.g. "*Theft rate per 100,000 by LGA, NSW 2022*"
- 5 Label the axis:** Chart Design → Add Chart Element → Axis Titles → label the horizontal axis with the unit.
- 6 Add a source note:** Insert → Text Box → place below the chart → type "Source: BOCSAR, [download date]".

Chart formatting rules

- ✓ Descriptive title (not "Chart 1")
- ✓ Axis labels with units
- ✓ Source and download date noted
- ✓ Bars sorted largest to smallest
- ✗ No 3D effects — ever
- ✗ No rainbow colours for one variable
- ✗ No decorative gridlines

Sorting your bars for impact

Sort your *data table* first (smallest to largest for horizontal bars — Excel reads bottom to top). The chart reorders automatically.

Exercise: Build two charts

Chart 1 — Bar chart (comparison)

- 1 Use your 5-LGA rate table. Select the LGA name and Rate columns (hold `Ctrl` to select both)
- 2 Insert → Clustered Bar. → *Your 5 LGAs should appear as horizontal bars*
- 3 Add a proper title, label the axis "Rate per 100,000 population", add a source note
- 4 Sort the data table so bars appear largest to smallest

Chart 2 — Line chart (trend)

- 1 Filter to **one LGA** and **one offence type**. Copy Year and Count to a fresh area.
- 2 Select both columns → Insert → Line chart
- 3 Title should include the LGA name, offence type, and year range
- 4 Look for the COVID dip around 2020–21. Add a text box annotation if you see it.

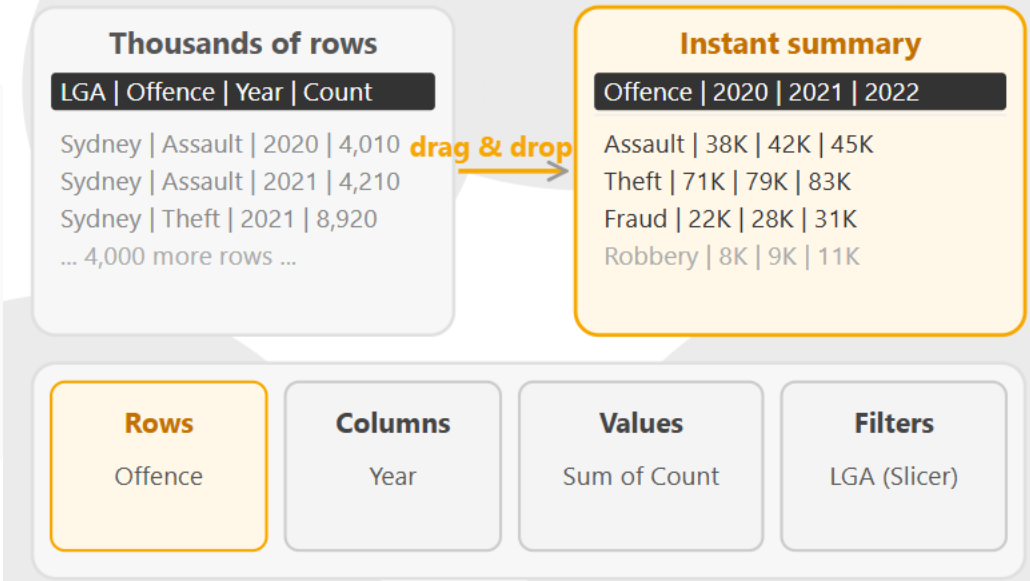


Module 6 — Pivot Tables

Add text

Pivot tables — your biggest time saver

A pivot table summarises thousands of rows into a neat cross-tab — in seconds, no formulas needed. You drag and drop fields to control the grouping.



The golden rule

Your data must have one row per observation, no merged cells, and no blank rows. Clean data first — then pivot.

When to use a pivot table

- You want totals by category — offence, state, year
- You want to compare two variables at once
- You want to quickly explore a large dataset
- You want to filter without using AutoFilter

Building your first pivot table

- 1 Click **anywhere inside your data**. One cell is enough — Excel detects the full range automatically.
- 2 Go to `Insert → PivotTable → New Worksheet → OK`. → *A new sheet opens with a blank pivot and the field pane on the right*
- 3 Drag **Offence Type** into the **Rows** box. → *Offence types appear down the left side*
- 4 Drag **Count** into the **Values** box. → *Totals appear. Which offence is highest?*
- 5 Drag **Year** into the **Columns** box. → *Full cross-tab by offence and year — done in about 30 seconds*
- 6 Try swapping fields around. Pivot tables are designed to be experimented with — you can't break them.

Adding a Slicer — visual filter

- 1 Click anywhere on your pivot table
 - 2 Go to `PivotTable Analyse → Insert Slicer` → tick **LGA** → OK
 - 3 Click any LGA name — the pivot table filters instantly. Hold `Ctrl` to select multiple. → *This is what slicers look like in a dashboard*
- 💡 Right-click a number in the Values area → **Show Values As** → **% of Grand Total** to switch from counts to percentages instantly — no formula needed.

Exercise: Build a pivot table from BOCSAR data

- 1 Go to your Working sheet. Click anywhere in your clean data → [Insert → PivotTable → New Worksheet → OK](#)
- 2 Drag **Offence Type** to Rows and **Count** to Values. → *What's the most common offence in NSW?*
- 3 Add **Year** to Columns. → *Which offence grew the most between the earliest and latest year?*
- 4 Add a **Slicer** for LGA → click Sydney. → *The whole table now shows Sydney only. Click Clear Filter to go back.*
- 5 Right-click a number → **Show Values As** → **% of Column Total**. What share of each year's total does Assault make up?
- 6 **Bonus:** With your pivot table selected → [Insert → PivotChart → Bar chart](#) . Add a proper title and source note.



Putting it all together — Youth justice in NSW

Build a one-page Excel summary using all three datasets — the kind of task analysts do in policy and government roles every day.



Step 1 — Get the data

Download youth detention rates from ROGS (Justice chapter) for NSW and 2 other states. Download youth offending data from BOCSAR.



Step 2 — Clean it

Paste into Excel. Check for text/number issues. Make LGA names and year labels consistent.



Step 3 — Calculate

Use ABS Census to find ATSI youth population. Calculate their share of detention vs share of total youth population.



Step 4 — Visualise

Create one bar chart (state comparison) and one line chart (trend over time). Format both with proper titles and sources.



Step 5 — Pivot

Build a pivot table summarising youth offence types for NSW over the last 5 years. Add a LGA slicer.



Step 6 — Narrate

Write 3 bullet points in a text box on the sheet explaining what the data shows and one key limitation.

What a good Excel data file looks like

✓ GOOD HABITS

- RAW sheet untouched — all edits on a Working copy
- Column headers clear, bold, in row 1
- Numbers right-aligned (stored as numbers, not text)
- No merged cells in the data area
- A cell records the source name and download date
- Charts have titles, axis labels, and source notes
- Formulas use cell references, not hardcoded numbers
- File saved as .xlsx, not .csv

✗ COMMON MISTAKES

- Editing raw data and losing the original
- Comparing raw counts across different-sized areas
- Charts titled "Chart 1"
- Rainbow colours for a single variable
- Pivot tables built on dirty data
- Treating correlation as causation in your write-up
- Not noting where data came from or when
- Sorting a selected column (scrambles the data)

The goal isn't a perfect spreadsheet — it's a **trustworthy** one. Someone else should be able to open your file, understand what you did, and reproduce it.

Exercise — Reading a table (ROGS)

- Open the education chapter and find the table showing government school recurrent expenditure per student by state. Answer: Which state spent the most per student? Which spent the least? What's the difference in dollars? *Skills: reading tables, identifying max/min, simple subtraction*
- *Difference between ROGS data tables and ROGS dataset:*
 - *Data Tables: have each table in a different tab in Excel*
 - *Dataset: has all the years and all the tables in 1 “sheet”*

Exercises

- **Exercise 2 — Spotting a trend (BOCSAR)** Download the NSW assault offence data. Plot total assault incidents per year over the last 10 years in a spreadsheet. Answer: Is the overall trend going up, down, or mixed? Were there any years that stood out as unusual? Why might that be? *Skills: creating a line chart, reading trends, thinking about context*
- **Exercise 3 — Comparing categories (BOCSAR)** Using the crime by offence type dataset, find the top 5 most common offences in NSW in the most recent year. Create a bar chart ranking them. *Skills: sorting data, bar chart creation, categorical comparison*
- **Exercise 4 — Rate vs count (ROGS)** Find the hospital waiting times or ambulance response data. Compare two states using both the raw number of responses and the rate per 100,000 population. Answer: Does the ranking change when you use rates instead of counts? Why does this matter? *Skills: understanding normalisation, rate calculation, critical thinking about data*
- **Exercise 5 — Geographic comparison (BOCSAR)** Download the LGA-level crime data. Find the 3 Local Government Areas in NSW with the highest and lowest rates of theft. Map them or list them in a table. *Skills: filtering, geographic thinking, rate interpretation*
- **Exercise 6 — Joining two datasets (ROGS + BOCSAR)** Find youth justice or juvenile detention figures from ROGS and cross-reference with BOCSAR youth crime data for NSW. Do states with higher youth crime rates also have higher detention rates? *Skills: combining datasets, correlation thinking, avoiding causal assumptions*
- **Exercise 7 — Telling a story (both sources)** Pick one topic — homelessness, mental health, or road safety — and find relevant data from both sources. Write a 3-sentence summary of what the data shows, suitable for a general audience. *Skills: synthesis, plain-language communication, identifying what matters*



Module 7 — Power BI In Excel

Add text

Power BI in Excel

- Power BI in Excel gives you capabilities that go beyond what native Excel can do, particularly around data scale, modeling, and interactivity.
- **Scale and performance** Native Excel struggles with large datasets — it's limited to ~1 million rows per sheet and slows down significantly before that. Power BI's in-memory engine can handle tens or hundreds of millions of rows efficiently, all within the Excel interface.
- **Data modeling** Excel's data tools are essentially single-table or simple relationship setups. Power BI brings a proper relational data model — you can connect multiple tables with defined relationships, creating a star or snowflake schema without flattening everything into one massive table.
- **DAX vs. formulas** Power BI uses DAX (Data Analysis Expressions), a formula language designed specifically for analytical calculations. DAX handles things like time intelligence, running totals, dynamic filtering, and complex aggregations that would require convoluted workarounds in standard Excel formulas or even Power Pivot.
- **Live connected reports** You can embed live Power BI reports directly in Excel, which means your visuals update from a centralized, governed dataset rather than a local file. This is particularly valuable in teams where you want a single source of truth.
- **Richer visualizations** Power BI's chart types and formatting options are more extensive than Excel's native charts, and they're interactive by default — clicking one visual filters others automatically.
- **When to stick with native Excel** For most everyday analysis — a few thousand rows, standard calculations, ad hoc work — native Excel is faster to build and easier to share. Power BI in Excel makes most sense when you're dealing with large data, complex models, or need to consume reports built by a BI team.

WORKSHOP COMPLETE

You've got the skills.
Now use real data.

ROGS, BOCSAR, and ABS all update regularly, are free to download, and cover topics that matter for NSW. Keep practising on data you care about.

pc.gov.au/rogs

bocsar.nsw.gov.au

abs.gov.au

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